Area Model Explorer

TEACHER KEY

5th Grade Math · Interactive Math · Explore & Reflect

Get started

Split rectangles into partial products to multiply two-digit numbers, then switch to fractions and multiply parts of a whole. Open the tool and use it to answer each question.

Explore

1. Show 36 × 24. List the four partial products.

600, 120, 120, 24

2. What is 36 × 24?

864

3. Show 45 × 13. Which partial product is the largest?

400 (40 × 10)

4. Explain why 48 × 25 is not just 40 × 20 + 8 × 5.

That leaves out 40 × 5 and 8 × 20; the full product is 1,200.

5. Switch to fractions. Show $\frac{2}{3} \times \frac{3}{4}$. What fraction is purple?

$\frac{6}{12} = \frac{1}{2}$

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Explore more

6. How many equal parts is the whole cut into for $\frac{2}{5} \times \frac{1}{2}$?

10

7. Find $\frac{3}{4} \times \frac{1}{2}$.

$\frac{3}{8}$

Reflect

8. Reflect: Why is $\frac{1}{2} \times \frac{1}{3}$ smaller than $\frac{1}{3}$?

You are taking half of one third, which is less than the whole third.

Teacher notes

Accept equivalent forms (for example 1/2 = 2/4 = 0.5) and any correct example where several answers are possible. Have students explain using the tool's picture.